

Project Proposal

Group #: Group 3

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Project Title: Smart Parking Access Control (SPAC)

Short Description:

Smart Parking Access Control (SPAC) proposes a computer vision system to support resident access decisions for a residential parking garage. The core system takes **images of incoming vehicles** as input. A licence plate detection model will be trained using publicly available annotated ALPR datasets to localize the plate region. Optical Character Recognition (OCR) will then be applied to extract the plate number. The recognized plate number is matched against a **simulated resident database (CSV)** to output an “Access Granted” or “Access Denied” decision.

The project focuses on building an end to end pipeline including dataset preparation, model training, evaluation, and inference. As an optional extension, the same pipeline can be applied to **frames extracted from recorded video**, treating each frame as an image for batch processing. Vehicle colour estimation may also be explored as a secondary verification signal to reduce false positives.

References/ Existing Implementations:

1. Silva, S. M., & Jung, C. R. (2018). License Plate Detection and Recognition in Unconstrained Scenarios. UFPR-ALPR Dataset. ([Paper \(IEEE\)](#) - [Dataset](#))
2. Xu, Z., Yang, W., Meng, A., Lu, N., Huang, H., Ying, C., & Huang, L. (2018). Towards End-to-End License Plate Detection and Recognition: A Large Dataset and Baseline. CCPD Dataset. ([Paper](#) - [GitHub](#))
3. Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You Only Look Once: Unified, Real-Time Object Detection. ([Link](#))
4. Jaidev AI. EasyOCR: Open-Source Optical Character Recognition Library. ([GitHub](#))

Project timeline:

Main components	Tentative Completion Date	Assigned to
1- Dataset Preparation & Problem Definition: Download and explore selected ALPR dataset (UFPR-ALPR / CCPD), prepare training/validation split, and design simulated resident database (CSV structure).	March 5	Arad & Sia
2- License Plate Detection Model Training: Implement and train a license plate detection model (YOLO-style architecture). Evaluate performance using metrics such as IoU and mAP.	March 15	Arad
3- OCR Integration & Plate Text Extraction: Integrate OCR module to extract plate numbers from detected regions. Clean and normalize recognized text. Evaluate recognition accuracy.	March 20	Sia
4- Database Matching & Access Decision Logic: Develop matching algorithm to compare recognized plate numbers with the simulated resident database. Implement Access Granted / Denied output logic.	March 24	Arad & Sia
5- End-to-End System Integration & Testing: Integrate detection, OCR, and matching into a complete pipeline. Perform system testing and error analysis on validation images.	March 28	Arad & Sia
6- Final Report, Demo & Presentation Preparation: Prepare final documentation, evaluation results, and presentation slides. Record system demonstration.	March 31	Arad & Sia